

Sai Teja Komuravelly | Data Engineer

ETL Pipelines | Spark & Databricks | AWS Cloud

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Professional Summary

- Data Engineer with **6+ years of experience** designing scalable data pipelines and distributed data processing systems across healthcare, banking, and fintech domains.
- Strong expertise in **Python, Spark, Scala, and advanced SQL**, building high-performance ETL pipelines for large-scale datasets exceeding **10TB+**.
- Proficient in **cloud-based data engineering on AWS and Databricks**, developing data lakes using S3, Glue, Athena, and Redshift for enterprise analytics.
- Skilled in **workflow orchestration using Apache Airflow** and real-time data processing with Kafka and Spark Streaming.
- Demonstrated expertise in **query optimization, data validation, pipeline monitoring, and cost-efficient cloud migration**, reducing ETL runtime by up to 40% and operational overhead by 50%.
- **Demonstrated effective collaboration** in Agile environments, mentoring engineers, and delivering BI-ready datasets for executive decision-making.

Professional Experience

Sr. Data Engineer | Modak Analytics

Sep 2023 – Current | IL, USA

Client: AbbVie

- Designed and developed scalable PySpark-based ETL pipelines to ingest data from multiple enterprise systems such as Salesforce, Veeva Vault, OLTP databases, and SFTP sources into an S3-based lakehouse platform.
- Built and maintained data transformation pipelines across medallion architecture to produce denormalized analytical datasets processing 100M+ records per batch used for reporting and downstream analytics.
- Implemented Apache Iceberg tables and Optimized Spark ETL jobs using partitioning strategies, broadcast joins, and shuffle tuning, improving pipeline runtime by 30–40% for large-scale transformation workflows.
- Conducted 50+ code reviews for PySpark pipelines, validating join conditions, merge logic, partitioning strategies, and primary key constraints to ensure scalable and maintainable ETL processes.
- Monitored and debugged production ETL pipelines, resolving failures caused by schema mismatches, upstream dependency issues, credential failures, and data quality inconsistencies.
- Collaborated with business stakeholders and cross-functional teams to gather requirements and convert them into JIRA stories aligned with Agile sprint cycles, ensuring timely delivery of data engineering solutions.
- Designed dataflows and pipeline dependencies across ingestion and transformation layers, ensuring proper execution sequencing of Raw, Silver, and Gold pipelines within the lakehouse architecture.
- Managed 15+ CI/CD deployments across Dev, Validation, and Production environments, executing change requests, validating data counts, and ensuring zero data loss during releases.
- Analyzed Spark job performance using Spark History Server, identifying bottlenecks such as skewed partitions and excessive shuffles to improve job efficiency and cluster utilization.
- Collaborated with validation teams to define 20+ test scenarios and edge cases, ensuring ETL pipelines meet enterprise data quality and compliance requirements.
- Managed cluster resource allocation and pipeline scheduling, ensuring optimal concurrency across Spark workloads without exceeding cluster memory or compute limits.
- Implemented DQ execution paths by redirecting duplicate records to DQ tables using spark configuration flags, preventing business pipeline downtime in 100% of duplicate scenarios.
- Authored and maintained Confluence design documents and JIRA user stories for new ingestion, TL/ATL transformations, and outbound workflows.

Data Engineer Intern | Prozech Solution

May 2023 – Aug 2023 | NC, USA

- Built an ETL pipeline for extracting data through web scraping API and orchestrated using airflow and get a notification to email once the airflow is done executing to process 140+ GB of data daily.
- Transformed transactional data to 30+ facts and dimensions tables utilizing Spark SQL, resulting in reduction in data redundancy through effective normalization.
- Collaborated with cross-functional teams to define and establish mapping documents for source to target mappings by analyzing the data and existing code.
- Achieved a 20% reduction in query response time by implementing optimized partitioning techniques and data distribution strategies in the Hive tables.

Client: Johnson and Johnson

- Leveraged Spark SQL and Airflow frameworks to design and deploy ETL pipelines, seamlessly integrating and processing 2+ TB of clinical trials data from 5 distinct sources (csv files, APIs, MySQL tables) enabling advanced research and analysis.
- Initiated API calls through postman to fetch all the responses in Json and load the Json's to staging layer in S3.
- Built scripts that perform end to end data cleansing process using SQL for 2 million records per migration removing data anomalies.
- Corrected the mapping of therapeutic area to disease for 5 data sources, resulting in 100% accuracy of mapping using Spark SQL.
- Worked on solutions for mastering of pharma industry Trials, Sites, and Investigators and increasing data accuracy by 85% by using TAMR MDM Tool and dedupe.
- Utilized Python to perform counterfactual analysis, filling missing values and achieving a 5% improvement in KPI Enrollment Duration resulting in improved overall performance of KPI accuracy.
- Standardized the 300K variations of 27K diseases for 5 data sources leveraging different ontologies using TERMite and CENTREE which resulted in an increase of data accuracy of diseases by 60%.

Jr. Data Engineer | Anvizon**Jun 2018 – Jul 2020 | India**

- Managed 50+ data analysis related requests from clients, traced issues, and proposed efficient resolution within 24 hours.
- Created 25+ data testing scenarios in the application to ensure claims data accuracy.
- Gathered detailed requirements from clients and compiled comprehensive documentation of functional and technical requirements, fostering project clarity.
- Reduced average search time to 30 MS by working on query optimization techniques using advanced SQL aggregation functions.

Skills

- **Programming & Scripting:** Python, Scala, SQL, Shell Scripting, Java, PySpark, Pandas, NumPy
- **Big Data Technologies:** Apache Spark, Hadoop, Hive, HBase, Kafka, Sqoop, Flume, HDFS, YARN, MapReduce
- **Cloud Platforms & Services:** AWS (S3, EMR, EC2, Lambda, Glue, Athena, RDS, Redshift, CloudWatch, Kinesis), Databricks, Azure Data Lake, Snowflake
- **Data Engineering & Architecture:** ETL Pipeline Development, Data Lake Architecture, Data Modeling, Data Warehousing, Data Migration, Data Partitioning, Data Quality Frameworks, Batch & Real-Time Data Processing
- **Workflow Orchestration & DevOps:** Apache Airflow, Luigi, Git, Bitbucket, Jenkins, CI/CD Pipelines, Docker, Kubernetes, JIRA, Agile/Scrum
- **Databases & Querying:** Teradata, MySQL, Amazon Redshift, PostgreSQL, SQL Server, Hive, NoSQL (MongoDB, Cassandra)
- **Analytics & Visualization:** Power BI, Tableau, Excel, Jupyter Notebook, Matplotlib, Seaborn
- **Monitoring, Testing & Operations:** Linux, Unix, MobaXterm, Postman, AWS CloudWatch, Log Analysis

Education

Master of Science in Data Analytics**Sep 2022 – Dec 2023**

Clark University, MA, USA

Achievements

- **"Client Delight" Award** – Recognized at ZS Associates for exceptional contributions to project delivery and stakeholder satisfaction.
- **SQL Lead for Data Analytics Workstream** – Served as the primary point of contact for SQL validation, preprocessing, and complex query optimization across the project.